MIL-F-20668
1 September 1962

MILITARY SPECIFICATION

FLANGES, IRON (MALLEABLE)

This specification has been approved by the Departments of the Army, the Navy, and the Air Force and supersedes the following specification:

Navy 45F6b
1 August 1934

This specification consists of this cover sheet and Navy Department Specification 45F6b, dated 1 August 1934, attached hereto, without modification:

Copies of this specification may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D.C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Depot, Scotia 2, New York. Both the title and identifying number or symbol should be stipulated when requesting copies.

When a request for this specification is received by a supplying activity, it will be necessary to attach this cover sheet to the pertinent specification before issue.

Custodians:
Army - Corps of Engineers
Navy - Bureau of Ships
Air Force

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45F6b
AUG. 1, 1934
SUPERSEDING
45F6a
Aug. 1, 1932

NAVY DEPARTMENT SPECIFICATION

FLANGES, IRON (MALLEABLE)

A. APPLICABLE SPECIFICATIONS

The following Navy Department specifications, of the issue in effect on date of invitation for bids, form a part of this specification, and bidders and contractors should provide themselves with the necessary copies:

General Specifications for Inspection of Material, together with Appendix II (Metals).
44T2, Threads, Standard, for Pipe and Pipe Fittings.

B. TYPES AND CLASSES

B-1. *Types.*—Malleable iron flanges shall be of the following types, as specified:

A. Threaded.
B. Threadless.

B-2. *Classes.*—Malleable iron flanges shall be of the following classes, as specified:

1. Finished.
2. Unfinished.

C. MATERIAL AND WORKMANSHIP

C-1. *Material.*—Malleable iron flanges shall be made of a quality of malleable iron which will resist distortion or cracking during the process of expanding steel tubing into the flange.

C-2. *Workmanship.*—Malleable iron flanges shall be of high-grade workmanship, and free from any defects which might affect their serviceability. They shall be sound, smoothly cored, true to form, free from cold shuts or injurious blowholes, and thoroughly cleaned of sand, scale, and all irregularities. All surfaces not machined shall be smooth and equal in this respect to the finish produced by pickling and tumbling.

D. GENERAL REQUIREMENTS

There are no general requirements applicable to this specification.

E. DETAIL REQUIREMENTS

E-1. *Shape and dimensions.*—Flanges shall conform in shape and dimensions to the respective sketch and dimensions herein. Where threadless flanges are to be used on ventilation and watertight piping

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systems, the internal diameter shall conform to the respective B and D dimensions herein, and where for use for drainage, fresh water, air escapes, and equalizing systems, the internal diameter shall conform to the respective B^1 and D^1 dimensions. (See par. H–2.)

E–2. *Finished flanges.*—Finished flanges shall be drilled. All surfaces designated "f" shall be machine finished. Threaded flanges shall conform to the respective sketch and dimensions herein. Threads shall be suitable for standard-weight iron pipe and shall conform to Navy Department Specification 44T2, referred to in section A.

E–3. *Unfinished flanges.*—In the case of these flanges there shall be sufficient material in excess of the dimensions specified to permit of machining to such dimensions, where finishing is required. Such flanges shall not be drilled or provided with the circular grooves or beveled edges on the inside, required in table.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

A representative number of flanges shall be taken from each lot offered for delivery and inspected for conformity with this specification.

G. PACKAGING, PACKING AND MARKING FOR SHIPMENT

G–1. *Packaging.*—Commercial packaging is acceptable under this specification.

G–2. *Packing.*—The subject commodity shall be delivered in standard commercial containers, so constructed as to insure acceptance by common or other carrier, for safe transportation, at the lowest rate, to the point of delivery.

G–3. *Marking.*—

G–3 (a). *Packages.*—Packages shall be marked with the name of the article, the type, the class, the size, and the quantity contained therein, as defined by the contract or order under which shipment is made, and the name of the manufacturer.

G–3 (b). *Shipping containers.*—Shipping containers shall be marked with the name of the article, the type, the class, the size, and the quantity contained therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

H. NOTES

H–1. It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

H–2. Requisitions for threadless flanges should state whether the flanges are for use on ventilation and watertight piping systems, or for drainage, fresh-water, air escapes, and equalizing systems. (See par. E–1.)

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H–3. Copies of Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington, D.C., except that naval activities should make application to the Commandant, Navy Yard, New York, N.Y. When requesting, refer to specification by both title and number.

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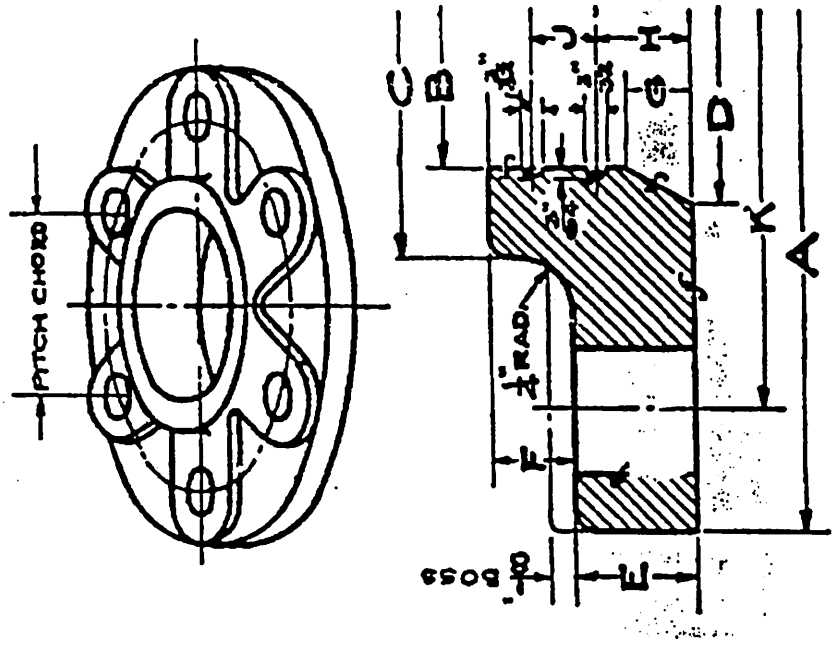
Threaded flanges

Size of pipe tap	Number of threads per inch	Diameter		Thickness		Bolts			
		Flange	Boss	Flange	Boss	Number	Diameter	Diameter of pitch circle	Pitch (chord)
		A	B	C	D			E	
Inches		*Inches*	*Inches*	*Inch*	*Inch*		*Inch*	*Inches*	*Inches*
1½	11½	5 1/16	2 3/4	1/2	5/8	6	½	3 1 1/16	1.97
2	11½	5 9/16	3 1/8	9/16	11/16	6	½	4 3/16	2.22
2½	8	6 1/4	3 5/8	9/16	1/2	6	½	5	2.50
3	8	6 5/8	4 1/4	9/16	9/16	8	½	5½	2.10
3½	8	7 9/16	4 13/16	9/16	5/8	8	½	6 1/16	2.32
4	8	7 11/16	5 5/16	9/16	5/8	8	½	6 9/16	2.51
4½	8	8 3/16	5 13/16	9/16	11/16	10	½	7 1/16	2.18
5	8	9 1/16	6 5/16	9/16	3/4	10	½	7 13/16	2.41
5½	8	9 3/16	6 3/4	5/8	3/4	10	5/8	8 5/16	2.57
6	8	10 1/8	7 1/8	11/16	3/4	12	5/8	8 7/8	2.30

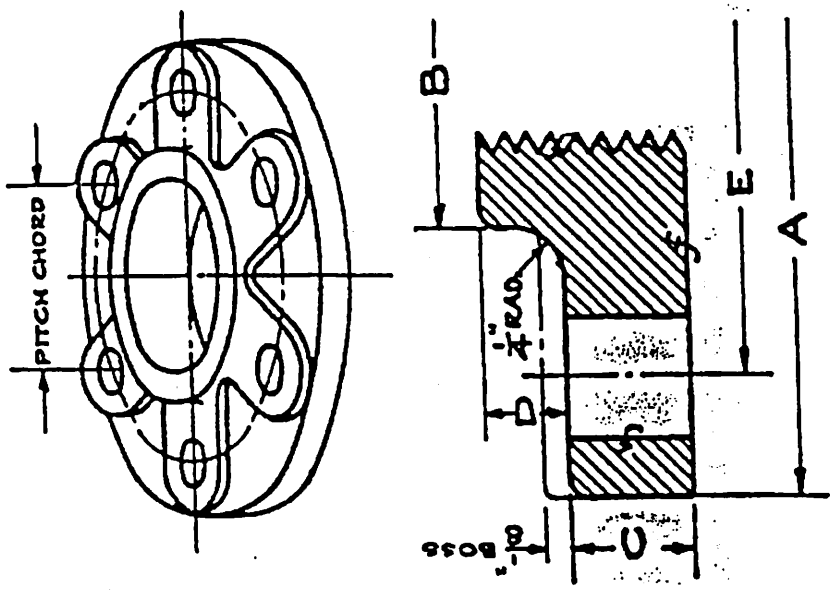
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THREADLESS FLANGES



THREADED FLANGE



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Threadless flanges

Nominal internal diameter	Diameter						Thickness					Bolts			
	Flange	Hole		Boss	Bevel		Flange	Boss	Depth of bevel	Groove—face to center line	Groove—center line to center line	Number	Diameter	Diameter of pitch circle	Pitch (chord)
	A	B	B¹	C	D	D¹	E	F	G	H	J			K	
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*		*Inch*	*Inches*	*Inches*
1½	5 1/16	2	2	2 5/16	2 3/16	2 3/16	7/16	5/16	1/4	3/8	1/4	6	1/2	3 13/16	1.97
2	5 5/16	2 1/4	2 1/2	3 1/16	2 7/16	2 1 3/16	7/16	5/16	1/4	3/8	1/4	6	1/2	4 1/16	2.22
2½	6 1 3/16	2 1/2	-------	3 3/16	2 1 3/16	-------	7/16	5/16	1/4	3/8	1/4	6	1/2	4 1/2	2.25
2½	6 1/4	2 3/4	3	3 1 1/16	2 1 3/16	3 3/16	1/2	5/16	1/4	3/8	5/16	6	1/2	5	2.50
2¾	6 3/8	3	-------	3 1 3/16	3 5/16	-------	1/2	5/16	1/4	3/8	5/16	6	1/2	5 1/4	2.63
3	6 3/8	3 1/4	3 1/2	4 1/16	3 1/16	3 1 1/16	1/2	5/16	1/4	3/8	5/16	8	1/2	5 1/2	2.10
3¼	6 7/8	3 1/2	-------	4 1/16	3 1 1/16	-------	1/2	5/16	1/4	3/8	5/16	8	1/2	5 3/4	2.20
3½	7 3/16	3 3/4	4	4 1/4	3 1 5/16	4 3/16	1/2	3/8	1/4	3/8	5/16	8	1/2	6 1/16	2.32
3¾	7 7/16	4	-------	5	4 3/16	-------	1/2	3/8	1/4	3/8	5/16	8	1/2	6 5/16	2.43
4	7 1 1/16	4 1/4	4 1/2	5 1/16	4 2 9/16	4 2 3/32	1/2	3/8	9/32	3/32	5/16	8	1/2	6 9/16	2.50
4¼	7 1 5/16	4 1/2	-------	5 1/4	4 1 3/16	-------	1/2	3/8	9/32	1 3/32	5/16	8	1/2	6 1 1/16	2.61
4½	8 1/16	4 3/4	5	5 3/8	4 1 3/16	5 7/32	9/16	3/8	5/16	7/16	5/16	10	1/2	7 1/16	2.18
4¾	8 1/16	5	-------	6	5 1 5/16	-------	9/16	3/8	5/16	7/16	5/16	10	1/2	7 5/16	2.26
5	9 1/16	5 1/4	5 1/2	6 5/16	5 3 1/16	5 3/8	9/16	3/8	2 1/16	7/16	5/16	10	5/8	7 13/16	2.41
5¼	9 5/16	5 1/2	-------	6 5/16	5 1 7/16	-------	9/16	3/8	2 1/16	7/16	5/16	10	5/8	8 1/16	2.49
5½	9 5/16	6	6	6 1 3/16	6 1 5/16	6 1/16	9/16	3/8	2 1/16	7/16	5/16	10	5/8	8 5/16	2.57
5¾	9 1 3/16	6	-------	7 1/16	6 1 5/16	-------	9/16	3/8	2 1/16	7/16	5/16	10	5/8	8 9/16	2.65
6	10 3/16	6 3/4	6 3/4	7 3/16	6 2 7/16	6 7/16	5/8	3/8	2 3/16	1 3/32	5/16	12	5/8	8 7/8	2.30
6¼	10 3/16	6 3/4	-------	7 5/16	6 2 7/16	-------	5/8	3/8	2 3/16	1 3/32	5/16	12	5/8	9 1/2	2.36
6½	10 5/16	7	7	7 7/16	7 1 7/16	7 5/32	5/8	3/8	3/8	1/2	5/16	12	5/8	9 5/8	2.43
7	11 1/16	7 3/4	7 3/4	8 5/16	7 3 9/16	7 2 9/32	5/8	3/8	1 3/32	1 7/32	5/16	12	5/8	10	2.59
7¼	11 1/16	7 3/4	-------	8 5/16	7 3 9/16	-------	5/8	3/8	1 3/32	1 7/32	5/16	12	5/8	10 1/4	2.66
7½	11 7/16	8 1/4	8	8 1 5/16	8 2 7/16	8 5/32	1 1/16	3/8	1 3/32	1 7/32	5/16	12	5/8	10 9/16	2.73
8	12 3/4	8 3/4	8 3/4	9 1/16	8 3 9/16	8 2 9/32	1 1/16	3/8	1 3/32	1 7/32	5/16	14	5/8	11 1/16	2.46
8½	12 1 5/16	9	9	10	9 1 9/16	9 9/32	1 1/16	3/8	1 5/32	1 7/32	5/16	14	5/8	11 5/8	2.59
9	13 1 5/16	10	10	10½	10 1 9/16	10 5/16	1 1/16	7/16	7/16	9/16	3/8	14	3/4	12 3/8	2.76

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